

Technology #162

Apply a heparin-coated polymer layer to the catheter lumen and integrate a thin piezoelectric strip that emits low-frequency ultrasonic waves (acoustic streaming) along the inner wall to synergistically inhibit clot adhesion.

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-09-02

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Apply a heparin-coated polymer layer to the catheter lumen and integrate a thin piezoelectric strip that emits low-frequency ultrasonic waves (acoustic streaming) along the inner wall to synergistically inhibit clot adhesion.

WHY DOES IT MATTER?

Documented problem 'catheter flow occlusion' concerning tunneled hemodialysis catheter. User-submitted engineering problem (session input): "Why do tunneled hemodialysis catheters lose flow patency within weeks of insertion despite flushing protocols?". [fda_maude] 100 records retrieved live (1992-2032). Illustrative records: 'MAUDE report 1000: Injury [SAME AS ABOVE]'; 'MAUDE report 100000: Injury [VITAL SIGNS]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2026-2026). Illustrative records: 'Commentary on: "clinical and haemodynamic evaluation of a novel physician modified inner branch iliac branch device for East Asians".'; 'Biosurfactants as Antibiofilm Agents for Medical Devices: Mechanisms, Evidence and Integration into Infection Prevention and Control.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 12 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in acoustic. 8-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 8-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Conduct a randomized bench■top flow loop study comparing (a) heparin■coated only catheters, (b) heparin■coated + ultrasonic actuation cat.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: phantom detection-threshold mapping vs depth and medium composition (measured quantity: detection sensitivity / false-alarm rate). Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: sound speed is medium- and temperature-dependent — source the value for the actual medium; alpha is frequency- and medium-dependent; two-way path doubles the exponent; oblique incidence and attenuation modify detectability; AC-002 (Wave speed relation) is applicable only under: non-dispersive propagation in the medium of interest.; AC-003 (Exponential attenuation) is applicable only under: homogeneous lossy medium; single frequency.; AC-001 (Acoustic impedance contrast (pulse-echo)) is applicable only under: normal incidence, plane wave, lossless media.; Model input variables requiring sourced values before any numeric result: c (m/s), f (Hz), A0 (-), alpha (1/m), d (m), Z1 (Rayl), Z2 (Rayl) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: lambda (m), A (-), R (-) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): operating frequency [f, Hz] value_status=UNKNOWN; element aperture [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; matching layer Z [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; excitation voltage and duty [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; detection threshold [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; coupling-path attenuation [alpha, 1/m] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Anticoagulant heparin coating on the catheter surface reduces thrombus formation, leading to longer failure-free survival.

POTENTIAL VALUE DRIVER: Addresses: Documented problem 'catheter flow occlusion' concerning tunneled hemodialysis catheter. User-submitted engineering problem (session input): "Why do tunneled hemodialysis catheters lose flow patency within weeks of insertion despite flushing protocols?". [fda_maude] 100 records retrieved live (1992-2032). Illustrative records: 'MAUDE report 1000: Injury [SAME AS ABOVE]'; 'MAUDE report 100000: Injury [VITAL SIGNS]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2026-2026). Illustrative records: 'Commentary on: "clinical and haemodynamic evaluation of a novel physician modified inner branch iliac branch device for East Asians".'; 'Biosurfactants as Antibiofilm Agents for Medical Devices: Mechanisms, Evidence and Integration into Infection Prevention and Control.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: acoustic

Disciplines: acoustics, transducer engineering, signal processing, materials science

Architecture: PROPOSED architecture (no physical system exists): transducer element + matching layer; excitation / receive electronics; signal chain (detection criterion); coupling path to target medium — realizing: Apply a heparin-coated polymer layer to the catheter lumen and integrate a thin piezoelectric strip that emits low-frequency ultrasonic waves (acoustic streaming) along the inner wall to synergistically inhibit clot adhesion.

Subsystems:

?: transducer element + matching layer - ?

?: excitation / receive electronics - ?

?: signal chain (detection criterion) - ?

?: coupling path to target medium - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Apply a heparin-coated polymer layer to the catheter lumen and integrate a thin piezoelectric strip that emits low-frequency ultrasonic waves (acoustic streaming) along the inner wall to synergistically inhibit clot adhesion.

Key Physics: Acoustics / ultrasonics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Acoustics / ultrasonics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

AC-002: Wave speed relation. Expression: $c = f * \lambda$. Applicability to this invention: APPLICABLE; holds under: non-dispersive propagation in the medium of interest. Assumptions: sound speed is medium- and temperature-dependent — source the value for the actual medium. Engaged variables: f .

AC-003: Exponential attenuation. Expression: $A = A_0 * \exp(-\alpha * d)$. Applicability to this invention: CONDITIONAL; holds under: homogeneous lossy medium; single frequency. Assumptions: α is frequency- and medium-dependent; two-way path doubles the exponent. Engaged variables: NOT ESTABLISHED.

AC-001: Acoustic impedance contrast (pulse-echo). Expression: $R = ((Z_2 - Z_1) / (Z_2 + Z_1))^2$. Applicability to this invention: CONDITIONAL; holds under: normal incidence, plane wave, lossless media. Assumptions: oblique incidence and attenuation modify detectability. Engaged variables: NOT ESTABLISHED.

Assumptions:

- sound speed is medium- and temperature-dependent — source the value for the actual medium
- α is frequency- and medium-dependent; two-way path doubles the exponent
- oblique incidence and attenuation modify detectability
- AC-002 (Wave speed relation) is applicable only under: non-dispersive propagation in the medium of interest.
- AC-003 (Exponential attenuation) is applicable only under: homogeneous lossy medium; single frequency.
- AC-001 (Acoustic impedance contrast (pulse-echo)) is applicable only under: normal incidence, plane wave, lossless media.
- Model input variables requiring sourced values before any numeric result: c (m/s), f (Hz), A_0 (-), α (1/m), d (m), Z_1 (Rayl), Z_2 (Rayl) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: λ (m), A (-), R (-) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): operating frequency [f , Hz] value_status=UNKNOWN; element aperture [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; matching layer Z [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; excitation voltage and duty [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; detection threshold [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN;

coupling-path attenuation [alpha, 1/m] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'catheter flow occlusion', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'detection threshold missed at depth (attenuation)', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'crosstalk between elements corrupting detection', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'resonance drift with temperature', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'bond-wire fatigue under cyclic excitation', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'coupling loss at interface deairing failure', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	maintain catheter flow patency for at least four weeks without increased infection risk	SOURCE_FACT
DI-002	Mechanism driver (proposed)	Anticoagulant heparin coating on the catheter surface reduces thrombus formation leading to longer failure-free survival	MODELLED
DI-003	Expected effect target	Sustained catheter patency for ≥ 4 weeks with a lower incidence of flow occlusion and no increase in infection risk.	MODELLED
DI-004	Domain design input pattern: targeted applicability	NOT_ESTABLISHED (candidate pattern; confirm against evidence)	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: allowed applicability (KATZ ET AL 1984)	NOT_ESTABLISHED (candidate pattern; confirm against evidence)	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: determined applicability	NOT_ESTABLISHED (candidate pattern; confirm against evidence)	ENGINEERING_PROPOSED
DI-007	Regulatory design input: compliance with standards	ISO 60601-2-32 (injection catheter) / UNKNOWN (heparin) / UNKNOWN (piezo) is never inferred	REGULATORY_PROPOSED
DI-008	Custodied observation: The value of transdermal heparin coating in preventing catheter thrombosis	https://pubmed.ncbi.nlm.nih.gov/24644694/ assessment hash: 0261ca008593578a6954c394a117600680463f2c951b5	CUSTODIED
DI-009	Custodied observation: Complications of heparin-coated catheters	https://pubmed.ncbi.nlm.nih.gov/24644694/ assessment hash: 1780da0f53501705a72024b19df5dc290495cce6aa1e	CUSTODIED
DI-010	Custodied observation: Four-Year Outcomes of Heparin-Coated Catheters	https://pubmed.ncbi.nlm.nih.gov/24644694/ assessment hash: 1780da0f53501705a72024b19df5dc290495cce6aa1e	CUSTODIED
DI-011	Custodied observation: Risk prediction for catheter thrombosis	https://pubmed.ncbi.nlm.nih.gov/24644694/ assessment hash: 1780da0f53501705a72024b19df5dc290495cce6aa1e	CUSTODIED
DI-012	Custodied observation: Heparin-Coated Catheter Thrombosis	https://pubmed.ncbi.nlm.nih.gov/24644694/ assessment hash: 1780da0f53501705a72024b19df5dc290495cce6aa1e	CUSTODIED

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	transducer element + matching layer realizing Apply a heparin-coated polymer layer to the catheter and integrate a thin piezoelectric strip that emits low	COMPLETED	details of the transducer definition
DO-002	excitation / receive electronics realizing Apply a heparin-coated polymer layer to the catheter and integrate a thin piezoelectric strip that emits low	COMPLETED	details of the electronics definition
DO-003	signal chain (detection criterion) realizing Apply a heparin-coated polymer layer to the catheter and integrate a thin piezoelectric strip that emits low	COMPLETED	details of the signal chain definition
DO-004	coupling path to target medium realizing Apply a heparin-coated polymer layer to the catheter and integrate a thin piezoelectric strip that emits low	COMPLETED	details of the coupling path definition
DO-005	transducer element + matching layer interfaces with excitation / receive electronics (all hands-off, to apply a heparin-coated polymer	CONCEPTUAL	signal chain / amplifier (all hands-off, to apply a heparin-coated polymer
DO-006	excitation / receive electronics interfaces with signal chain (detection criterion): signal chain / amplifier (all hands-off, to apply a heparin-coated polymer	CONCEPTUAL	signal chain / amplifier (all hands-off, to apply a heparin-coated polymer
DO-007	signal chain (detection criterion) interfaces with coupling path to target medium: signal chain / amplifier (all hands-off, to apply a heparin-coated polymer	CONCEPTUAL	signal chain / amplifier (all hands-off, to apply a heparin-coated polymer
DO-008	Device boundaries: patient/fluid inlet and outlet terminal control/data points required by applying heparin-coated polymer layer to the catheter	CONCEPTUAL	control/data points required by applying heparin-coated polymer layer to the catheter
DO-009	Flow/force/signal path geometry realizing the mechanism is known heparin-coated polymer layer to the catheter and integrate a thin piezoelectric	UNKNOWN	dimensions of the catheter
DO-010	Operating range for operating frequency within which the mechanism holds its stated behavior computed bounds for operating frequency; verification of range	UNKNOWN	stated behavior
DO-011	Operating range for element aperture within which the mechanism holds its stated behavior computed bounds for element aperture; verification of range	UNKNOWN	stated behavior
DO-012	Operating range for matching layer Z within which the mechanism holds its stated behavior computed bounds for matching layer Z; verification of range	UNKNOWN	stated behavior
DO-013	Operating range for excitation voltage and duty within which the mechanism holds its stated behavior computed bounds for excitation voltage and duty; verification	UNKNOWN	stated behavior
DO-014	Operating range for detection threshold within which the mechanism holds its stated behavior computed bounds for detection threshold; verification of range	UNKNOWN	stated behavior
DO-015	Operating range for coupling-path attenuation within which the mechanism holds its stated behavior computed bounds for coupling-path attenuation; verification	UNKNOWN	stated behavior
DO-016	No closed-loop control is proposed: the invention operates as an open-loop as specified; this record exists so the omission is explicit and auditable	CONCEPTUAL	open-loop as specified
DO-017	Information path: sensing element -> decision/monitor control (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in	CONCEPTUAL	act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in
DO-018	Fixture enabling: impedance-contrast detection threshold concept named fixture enabling; acceptance instrumentation	UNKNOWN	fixture enabling; acceptance instrumentation
DO-019	Fixture enabling: crosstalk and noise floor characterization concept named fixture enabling; acceptance instrumentation	UNKNOWN	fixture enabling; acceptance instrumentation
DO-020	Fixture enabling: durability under cyclic excitation — rig concept named fixture enabling; acceptance instrumentation	UNKNOWN	fixture enabling; acceptance instrumentation

ID	Description	Status	Missing Inputs
DO-021	Fixture enabling: phantom detection-threshold mapping	UNKNOWN - rig concept framed, guidelines, instrumentation	UNKNOWN - rig concept framed, guidelines, instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
operating frequency	UNKNOWN (no sourced value)	Hz	ENGINEERING_PROPOSED	(design assigned and target) value and verification method
element aperture	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned and target) value and verification method
matching layer Z	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned and target) value and verification method
excitation voltage and duty	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned and target) value and verification method
detection threshold	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned and target) value and verification method
coupling-path attenuation	UNKNOWN (no sourced value)	1/m	ENGINEERING_PROPOSED	(design assigned and target) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
catheter flow occlusion	Fails by catheter flow occlusion	design electrode NOT in STA but in VBB (requires design gain)	MCNOWN the untested condition failed in the (Prob)	
detection threshold	Fails by depth (attenuation) appeared at depth category above F0.05 say attempt budget		delay in the mesh in the leg, clamping in the ventral (Athe)	
crosstalk between electrodes	Fails by crosstalk between elements isolating (deflecting) F0.05 Bica de coupling		delay in the mesh in the leg, clamping in the ventral (Athe)	
resonance drift with temperature	Fails by resonance drift with temperature	temperature in the pipe at 1500 Hz and vibration	delay in the mesh in the leg, clamping in the ventral (Athe)	
bond-wire fatigue under fatigue	Fails by bond wire fatigue	undergo to a high rate of 5000 cycles limits	delay in the mesh in the leg, clamping in the ventral (Athe)	
coupling loss at interface	Fails by coupling loss at interface	coupling in the interface and mate	delay in the mesh in the leg, clamping in the ventral (Athe)	

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
catheter flow occlusion	Fails by catheter flow occlusion: the device fails to maintain flow	design (location of NOT ESTABLISHED)	residual uncertainty: spreading data
detection threshold missed	Fails by detection threshold missed at depth (at end of frequency selection)	depth (at end of frequency selection)	depth (at end of frequency selection)
crosstalk between elements	Fails by crosstalk between elements coupling detection (acoustic)	electrical coupling (acoustic)	electrical coupling (acoustic)
resonance drift with temperature	Fails by resonance drift with temperature	acoustic coupling (acoustic)	acoustic coupling (acoustic)
bond-wire fatigue under cyclic excitation	Fails by bond-wire fatigue under cyclic excitation	excitation (acoustic)	excitation (acoustic)
coupling loss at interface	Fails by coupling loss at interface	coupling (acoustic)	coupling (acoustic)

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Conduct a randomized bench	top flow	Complete and pairing (a)	Complete and pairing (a)
VF-002	run killer experiment: falsification_test	run killer experiment (a)	run killer experiment (a)	run killer experiment (a)
VF-003	phantom detection-threshold mapping	phantom detection-threshold mapping	phantom detection-threshold mapping	phantom detection-threshold mapping
VF-004	resonance tracking across the temperature	resonance tracking across the temperature	resonance tracking across the temperature	resonance tracking across the temperature
VF-005	accelerated cyclic-excitation run-out with	accelerated cyclic-excitation run-out with	accelerated cyclic-excitation run-out with	accelerated cyclic-excitation run-out with
VF-006	transmitted-energy check with deliberate	transmitted-energy check with deliberate	transmitted-energy check with deliberate	transmitted-energy check with deliberate

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	PZT / PVDF active element	standard ultrasonic transducers	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
PZT / PVDF active element	domain candidate material	1	NOT ESTABLISHED	PZT / PVDF active element	NOT ESTABLISHED	NOT ESTABLISHED	Reliability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- diced-ceramic + backing assembly: domain registry manufacturing route - candidate only; no process qualification exists
- matching-layer lamination and bonding: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-002', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter operating frequency [f Hz]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter element aperture [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter matching layer Z [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter excitation voltage and duty [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter detection threshold [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter coupling-path attenuation [alpha 1/m]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}

- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}
- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-006', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-006'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	maintaining the proposed physical verification	1050 Hz frequency	ENGINEERING_PROPOSED	1050 Hz frequency	1050 Hz frequency
WP-02	constraint-compliance envelope article	per method	constraint-compliance	NOT ESTABLISHED	constraint-compliance	NOT ESTABLISHED (report quoted)
WP-03	article for VF-001	per method	Conduct a randomized	NOT ESTABLISHED	Study 1 configuration	NOT ESTABLISHED (try be quoted) (b) he
WP-04	article for VF-002	per method	run killer experiment:	NOT ESTABLISHED	Vendor data	NOT ESTABLISHED (1575382412046) quoted)
WP-05	article for VF-003	per method	phantom detection threshold	NOT ESTABLISHED	VF-003 verification	NOT ESTABLISHED (to be quoted) detection
WP-06	article for VF-004	per method	resonance tracking	NOT ESTABLISHED	VF-004 verification	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-005	per method	accelerated cyclic-exon	NOT ESTABLISHED	VF-005 verification	NOT ESTABLISHED (to be quoted) stress
WP-08	article for VF-006	per method	transmitted-energy ch	NOT ESTABLISHED	Defining challenge	NOT ESTABLISHED (to be quoted) measured q

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 8-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.